Lesson notes · for you

Nearest neighbours

Data science · Level 3 · Lesson 3

Time
30 min

Materials
A notebook and a pencil; A kitchen scale and some fruit (no scale: measure each fruit's length with a ruler instead); A phone or tablet with Darsi, for the lab

Learning goals

- Tell labels from numbers.
- Classify by nearest neighbours.
- Choose k sensibly.

Activity: The mystery fruit

1. Warm-up (5 min): Hear a fruit's weight and colour score and guess what it is.
2. Measure (10 min): With a partner, weigh fruit, score colour and add dots to the board.
3. Classify (10 min): Find the nearest neighbours of each mystery fruit with k = 1 and k = 3.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- When did k change the answer?
- Is a fruit type a number or a label?
- Why rescale the grams?

Check yourself

- Neighbours: blue, red, blue, green, blue. With k = 5, which label wins?

Try it this week
Weigh some fruit at home, give each colour a score from 1 to 5 and plot them on a square chart, each axis running from its smallest value to its largest. Then ask someone to bring a mystery fruit and find its three nearest neighbours.

Worksheet

Nearest neighbours

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Name _____ Class _____ Date _____

1 **A model chooses a reading group for a new pupil. Does it predict a number or a label?**

◯ A label ◯ A number ◯ Neither

2 **k-NN with k = 1 can overfit.**

◯ True ◯ False

3 **Classify the fruit at (5, 7) with 1-NN and print its label.**

```
pts = [(2, 3, "lime"),
       (6, 8, "mango"),
       (9, 2, "apple")]
best = pts[0]
print(best[2])
```

Write your program:

4 **Weight is in grams (100–300) and colour is a score 1–5. What should you do before k-NN?**

◯ Drop the colour ◯ Use k = 1 ◯ Rescale both features to 0–1

5 **Which of these tasks predict a number, so they are regression? Choose all that apply.**

☐ Predict tomorrow's temperature ☐ Predict a house price

☐ Choose a reading group for a new pupil ☐ Spam or not spam

✎ **Where do apps sort things by comparing them with similar things? What could go wrong?**

Check your answers

Nearest neighbours

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1 **A label**

A reading group is a label, so this is classification.

2 **True**

A single odd neighbour decides, so noise is followed.

3

```
pts = [(2, 3, "lime"),
       (6, 8, "mango"),
       (9, 2, "apple")]
best, bd = pts[0], 1e9
for p in pts:
    dx, dy = p[0] - 5, p[1] - 7
    d = (dx ** 2 + dy ** 2) ** 0.5
    if d < bd:
        best, bd = p, d
print(best[2])
```

(6, 8) is about 1.4 away, much closer than the others, so 1-NN predicts mango.

4 **Rescale both features to 0–1**

Otherwise grams dominate every distance.

5 **Predict tomorrow's temperature; Predict a house price**

Temperature and price are numbers, so regression; spam and a reading group are labels, so classification.